1) $R(T) = R_0 + \alpha \Delta T$

$\Delta T = T - T_0$

a) Spænningsdeling

$$U_{out} = U_0 \left(\frac{R_2}{R_1 + R_2} - \frac{R_3}{R_3 + R(T)} \right)$$
$$= U_0 \left(\frac{R_2}{R_1 + R_2} - \frac{P}{R_3 + R_0 + \alpha \Delta T} \right)$$

b) For å få en lineær tilnærming

$\alpha \Delta T << R_0$

$$U_{out} \approx U_0 \frac{F}{(F+1)^2} \frac{\alpha \Delta T}{R_0}$$

$$F = \frac{R_2}{R_1} = \frac{R_0}{R_0}$$

Fra a)

$$U_{out} = U_0 \left[\frac{1}{1 + \frac{R_1}{R_2}} - \frac{1}{1 + \frac{R_3}{R_2} + \frac{\alpha \Delta T}{R_2}} \right]$$
$$= U_0 \left[\frac{1}{F+1} - \frac{1}{1 + F + \frac{\alpha \Delta T}{R_0}} \right]$$
$$= U_0 \frac{1}{F+1} \left[1 - \frac{1}{1 + \frac{\alpha \Delta T}{R_0 (F+1)}} \right]$$
$$= \frac{U_0}{F+1} \left(1 - \frac{\alpha \Delta T}{R_0 \frac{F+1}{F}} \right)$$
$$= U_0 \frac{F}{(F+1)^2} \frac{\alpha \Delta T}{R_0}$$

c) $U'(F) = \frac{1-F}{(F+1)^3}$ $F=1$ so $U'(F)=0$

$F=1$ er maksimum

$$U_{out} = U_0 \frac{1}{4} \frac{\alpha \Delta T}{R_0}$$

Når $R(T)$ endres 0%,

vil U_{out} 0.125% av U_0

$$U''(F) = \frac{2(F-1)}{(F+1)^4}$$

d) $$U_{out} = A_1 \left(\frac{\alpha \Delta T}{R_0} \right) + A_2 \left(\frac{\alpha \Delta T}{R_0} \right)^2 + \dots$$

$$A_1 = U_0 \frac{F}{(F+1)^2}$$

$$A_2 = -U_0 \frac{F-2}{(F+1)^3} \qquad A_2 = -\frac{U_0}{8}$$

2) $R_{20} = R_{20,0}$

$R = R_0$

$V_{out} = 0$

$R_{20}: R_{20,0} \to R_{20,0} + \Delta R_{20,T} + \Delta R_{20,\varepsilon}$

$R: R_0 \to R_0 + \Delta R_T$

Sperare Torr pert
on i gi gemons
pol tenile